

Visual Studio Code interface showing a Python application (app.py) and its deployment to a Kubernetes cluster using Docker and Kubernetes.

EXPLORER

- FLASK-K8S-DEMO
 - app.py
 - requirements.txt
 - Dockerfile
 - deployment.yaml
 - service.yaml

app.py

```
1 from flask import Flask, jsonify
2
3 app = Flask(__name__)
4
5 @app.route('/')
6 def home():
7     return jsonify({
8         "message": "Hello from Flask API!",
9         "status": "success"
10    })
11
12 @app.route('/api/data')
13 def get_data():
14     data = {
15         "id": 1,
16         "name": "Docker Kubernetes Demo",
17         "technology": "Flask + Docker + Kubernetes"
18     }
19     return jsonify(data)
20
21 @app.route('/api/health')
22 def health():
23     return jsonify({"status": "healthy"})
24
25 if __name__ == '__main__':
26     app.run(host="0.0.0.0", port=5000)
```

TERMINAL

```
PS C:\Users\sahhانا\flask-k8s-demo> docker build -t sahhanaa44/flask-k8s-demo:1.0 .
[*] Building 11.2s (9/9) FINISHED
-> [internal] load build definition from Dockerfile
-> --> transferring dockerfile: 139B
-> [internal] load .dockerignore
-> --> transferring context: 2B
-> [internal] load metadata for docker.io/library/python:3.10-slim
-> [1/4] FROM docker.io/library/python:3.10-slim@sha256:6d...
-> [2/4] WORKDIR /app
-> [3/4] COPY requirements.txt .
-> [4/4] RUN pip install --no-cache-dir -r requirements.txt
-> exporting to image
-> exporting layers
-> writing image sha256:3f2b8e7d9e6bdc3f8a1d212123...
-> naming to docker.io/sahhanaa44/flask-k8s-demo:1.0
```

PROBLEMS

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
sahhanaa44/flask-k8s-demo	1.0	3f2b8e7d9e6b	2 minutes ago	168MB
python	3.10-slim	b0a9641ffef1	2 weeks ago	124MB

PORTS

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/flask-k8s-service	NodePort	10.100.152.12	<none>	80:30880/TCP	2m45s

DEPLOYMENT

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/flask-k8s-demo	2/2	2	2	2m45s

REPLICASETS

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/flask-k8s-demo-5b6d5f7f8c	2	2	2	2m45s

PODS

NAME	READY	STATUS	RESTARTS	AGE
pod/flask-k8s-demo-5b6d5f7f8c-abc12	1/1	Running	0	2m45s
pod/flask-k8s-demo-5b6d5f7f8c-def34	1/1	Running	0	2m45s

TERMINAL OUTPUT

```
PS C:\Users\sahhانا\flask-k8s-demo> curl http://localhost:5000/
{"message": "Hello from Flask API!", "status": "success"}

PS C:\Users\sahhانا\flask-k8s-demo> curl http://localhost:5000/api/data
{"id": 1, "name": "Docker Kubernetes Demo", "technology": "Flask + Docker + Kubernetes"}

PS C:\Users\sahhانا\flask-k8s-demo> curl http://localhost:5000/api/health
{"status": "healthy"}

PS C:\Users\sahhانا\flask-k8s-demo>
```